Master Learning Plan v9

Section-by-section: v6 floor honored, depth added, resources expanded

Medical-Imaging + Geometric/Molecular-ML — Research-Engineer Readiness — Master Learning Plan (v9)

Owner: MD Hashim · **Window:** opens now (mid-Aug 2026) · **Timeline:** v6-aligned — Block 1 core ~Mar 2027, Block 2 ~Jul 2027, then expertise grinding (each block may run ~1 month longer for the added depth — pre-approved) · **Total effort:** ~2,390 focused hours · **Structure:** a job-critical spine (Blocks 1–2) + a full-depth **Quantum/QML track (Phase Q)** + three robust continuous threads (DSA/C++, implement-by-hand ML, research output) + the **Confusion Buffer** scaffold + an application overlay that starts **now**.

> **v9 is v6 restored as the floor, then deepened.** Per your instruction, **no section is below its v6 hours** — the CV-expert block and quantum are back at full strength, and I've **added new resources and practice** to the sections earlier drafts had trimmed. On top of that floor, v9 *increases* time where more depth clearly pays off for you (math, fundamentals, DL theory, the geometric/molecular spike, and the QML track), and it makes **Threads T, M, R robust enough to walk into any interview confident** in DSA, from-scratch implementation, and research discussion. All the *sequencing and positioning* wins from v7/v8 are kept — Research-Engineer as the near-term target (RS is PhD-gated, long-term), the spike protected, capstones renumbered, and the **Confusion Buffer** (your shock absorber for hard terminology) — but none of them as hour cuts. Because the plan is incremental, you can **start applying and interviewing after ~1–2 months** of fundamentals brush-up and climb the tiers as each capstone lands. All v6 URLs are preserved; new ones are added and marked.

Changelog (v8 → v9) — read this first

Area	v8	v9
Hours floor	Some sections cut below v6 (CV-expert → 30; threads trimmed; C5 dropped)	**No section below v6.** CV-expert restored to full (245), quantum full (305), C5 (deploy) restored, threads increased
Added resources	—	**New resources + practice added** to the previously-trimmed sections (CV-expert, quantum, DSA) — see each section and the index
Total hours	~1,870	**~2,390** (v6 floor + depth increases + robust threads)
Threads T / M / R	2.5 / 2 / 1.5 hrs/wk	**5 / 3 / 2 hrs/wk** — robust, so no interview area feels shaky; plus a fuller Phase-9 crescendo
Timeline	Understanding-gated, no calendar	**v6-aligned calendar targets** (Block 1 core ~Mar, Block 2 ~Jul) + self-tests as the real gates + pre-approved ~1-mo/block extension
Phase 8 (CV expert)	Cut to 30 (literacy)	**Restored to 245** (all of v6's classical/geometric/3D/video/edge) **+ new resources**; video sub-blocks can *optionally* be pulled early to unlock Tier-1 sooner (an option, not a cut)
Quantum (Phase Q)	305, off-path only	**305 + added resources**; **two sequencing options** (integrated into Block 1 like v6, *or* parallel/after-Tier-2) — you choose
Depth increases (kept from v8)	P0 55, P1 205, P3 160, P6 185	**Kept and nudged up** where useful (P1 210, P3 165, P6 185, P2 195)
Confusion Buffer, QML branch, PhD/pub/risk, RE-target, capstone renumbering	Added	**All kept**

The program at a glance

Job-critical spine — Blocks 1–2 (~1,535 hrs).

- *Block 1 — Foundations, domain & the spike (~975 hrs core; target ~Mar–Apr 2027):* a thorough fundamentals rebuild → a full, intuition-first math build → your medical-imaging domain (full v6 depth) → DL theory & modeling → production systems → the **geometric/molecular-ML spike** (the differentiator for the relocation target).
- *Block 2 — Breadth & lab-interview readiness (~560 hrs + interviewing; target ~Jul–Aug 2027):* GenAI & agents (domain-focused) → the **full CV-expert block** (classical/geometric/3D/video/edge — restored) → a research + big-tech interview crescendo.

Quantum/QML track — Phase Q (~305 hrs, full depth). A complete, intuition-first quantum + QML build from near-zero — because you want the expertise. **Two ways to run it:** (A) integrated into Block 1 like v6 (finishes ~Mar with everything; heavier weekly load), or (B) parallel/after-Tier-2 (interview sooner, grind quantum alongside/after — the default, best for early value). Full hours either way. It seeds a QML career branch (QpiAI) and deepens the molecular spike's "quantum-informed" story.

Threads — robust (from the Phase-1 self-test gate). DSA + C++ (~5 hrs/wk), implement-by-hand ML (~3 hrs/wk — the highest-ROI thread and a concept-cementer), research output & networking (~2 hrs/wk). Sized so you're confident in every interview surface. They flex down only on consolidation/heavy weeks.

The Confusion Buffer. Your shock absorber for confusing terminology — intuition-first ordering, a living glossary, spaced re-derivation, a Feynman gate, an unblock ritual, consolidation weeks, understanding-gated checkpoints, and multi-teacher triangulation for the hardest topics.

Application overlay (from now). Your applied medical-imaging skill makes you eligible at Tier 0 today. Apply after ~1–2 months of fundamentals brush-up and climb the tiers as each capstone lands.

Why this order — early value + full depth (both, not either/or)

You named the tension exactly: a **vanilla model config hits a ceiling**, and you can't push past baseline — or explain *why* a model behaves as it does — without solid math and fundamentals. So v9 gives the foundations the generous, intuition-first time they need for *you*, restores full breadth everywhere so no interview area feels shaky, **and** preserves early value three ways that don't require finishing the learning first:

1. **Apply from ~month 2 in parallel.** Your applied imaging skill (nnU-Net/MONAI, MRI/fluorescence) is already hireable at Tier 0; early loops pay you in feedback and calibration.
2. **Each capstone unlocks a higher tier.** C1 (seg) → Tier 0/1; C2 (video) → Tier 1; C3 (molecular spike) → Tier 2; the crescendo → Tier 3. You climb incrementally as depth matures.
3. **Quantum can stay off the job-critical path** (option B) so full QML depth never delays your ladder — or fold it into Block 1 (option A) if you want it done by March like v6.

Pace & timeline (v6-aligned; self-tests are the real gates)

Calendar targets mirror v6; the **phase self-tests and Feynman gates are the true gates** — advance on understanding, not dates. Given the added depth and your slow-intuitive style, each block may run ~1 month longer, which you've pre-approved. A steady ~30–38 hrs/wk on phases + ~10 hrs/wk on threads (flexing down on heavy weeks) hits these targets:

Milestone	Focused hrs	Target date (v6-aligned)
Fundamentals brushed up → start applying	P0 + early P1 (~120)	~Oct 2026 (month ~2)
Tier-0 interview-ready (+ Capstone 1)	~360 core	~Nov–Dec 2026
Tier-1 strengthened (+ Capstone 2 video)	~+ video block	~Jan–Feb 2027
Block 1 core complete → Tier-2 RE credible (+ spike + Capstone 3)	~975 core	**~Mar–Apr 2027**
Block 2 complete → Tier-3 loops (CV-expert + crescendo)	~1,535 spine	**~Jul–Aug 2027**
Quantum/QML track (Phase Q)	+~305	parallel throughout, or a concentrated block ~Aug–Nov 2027 (option B); or folded into Block 1 (option A)
Expertise grinding + active interviewing	ongoing	~mid-2027 → end-2027

Critical path to Tier-2 RE (relocation target): P0 → P1 (full) → P3 → **P6 spike + Capstone 3** → P5 ML-system-design slice → P9 job-talk + implement-from-scratch, with **Thread M running throughout**. Quantum (Phase Q, option B), the full CV-expert block, and GenAI are *not* on this path.

The Confusion Buffer — your shock absorber for hard concepts

Apply this to every topic that feels slippery. It turns "I keep forgetting the terminology / I'm not intuitive about this" into a managed system. Use it constantly — it's as important as any phase.

1. **Intuition-first ordering (never start rigorous cold).** For each hard topic: **(a) visual/intuitive explainer first** (3Blue1Brown, StatQuest, a good blog) → **(b) the rigorous source** (MIT/Strang, a paper) → **(c) implement or derive it by hand.** Starting rigorous before you have the picture is the #1 cause of slow-learner grind.
2. **Living glossary ("terminology war chest").** One doc. Every confusing term: a one-line plain-English definition, a tiny concrete example, and where it shows up. Add every session. Fixes "forget most of the terminology" — you look terms up in *your own words*.
3. **Spaced re-derivation (Anki-style, for concepts).** Anything that felt hard re-explained/re-derived at **1 day → 3 days → 1 week → 1 month.** Anki https://apps.ankiweb.net/ or a dated checklist.
4. **The Feynman gate (the real "done" test).** Not done until you can explain it out loud, plain language, **to a 12-year-old, without notes.** If you can't, you've found the exact gap — go back to step 1 for that piece. (This *is* your interview-articulation training.)
5. **The unblock ritual (no grinding).** Stuck >30 min: log the exact confusion, then **switch teacher/modality** rather than re-reading the same source. A different angle usually unlocks it fast.
6. **Consolidation weeks (schedule-level shock absorber).** Every ~4–5 weeks: **no new material** — only re-derive, review the glossary, re-implement shaky concepts. Bank these liberally.
7. **Understanding-gated checkpoints.** Self-tests are pass/fail on *explain + derive + implement unaided*. **Do not advance until you pass.**
8. **Multi-teacher triangulation for the hardest ~6 topics.** Backprop, attention & $\sqrt{d_k}$, SVD/eigendecomposition, diffusion/score-matching, equivariance, and VQE/barren plateaus each get **2–3 different teachers** (below). If one doesn't click, another will.

Triangulation sources: Backprop — 3B1B NN https://www.youtube.com/playlist?list=PLZHQObOWTQDNU6R1_67000Dx_ZCJB-3pi · micrograd https://github.com/karpathy/micrograd · d2l.ai https://d2l.ai/ ; Attention/$\sqrt{d_k}$ — Annotated Transformer http://nlp.seas.harvard.edu/annotated-transformer/ · 3B1B https://www.youtube.com/@3blue1brown · Lilian Weng https://lilianweng.github.io/ ; SVD/eigen — 3B1B LA https://www.youtube.com/playlist?list=PLZHQObOWTQDPD3MizzM2xVFitgF8hE_ab · MIT 18.06 https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/ · MIT 18.065 https://ocw.mit.edu/courses/18-065-matrix-methods-in-data-analysis-signal-processing-and-machine-learning-spring-2018/ ; Diffusion — Yang Song https://yang-song.net/blog/2021/score/ · Lilian Weng https://lilianweng.github.io/posts/2021-07-11-diffusion-models/ · CS236 https://www.youtube.com/playlist?list=PLoROMvodv4rPOWA-omMM6STXaWW4FvJT8 ; Equivariance — Bronstein https://geometricdeeplearning.com/lectures/ · e3nn tutorial https://blondegeek.github.io/e3nn_tutorial/ · Veličković https://www.youtube.com/results?search_query=petar+velickovic+gnn+lectures ; VQE/barren plateaus — PennyLane demos https://pennylane.ai/qml/demos_quantum-chemistry · Quantum Country https://quantum.country/qcvc · Schuld https://www.youtube.com/results?search_query=maria+schuld+taking+stock+quantum+machine+learning

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Threads (robust muscle-memory + output scaffold)

*All three start at the **Phase-1 self-test** (when you can derive backprop + attention unaided) and run continuously. Do them first thing, before the day's phase work. Sized for genuine expertise and interview confidence; flex down to ~2–3 hrs combined only on consolidation/heavy weeks.*

Thread T — DSA + C++ · ~5 hrs/week *(up from v6's 4)* · so you never feel shaky on coding rounds

RE loops have two medium-hard coding rounds; this thread builds real fluency, not just survival. Sustained ~5 hrs/wk over the window ≈ 250–280 hrs — genuinely strong.

Stage T1 — C++ fundamentals (Weeks ~1–8 of the thread, ~40 hrs): types, references/pointers, RAII, the STL, classes, templates, move semantics, smart pointers.

- learncpp.com (sections 1–17) — https://www.learncpp.com/ · The Cherno — https://www.youtube.com/@TheCherno · cppreference — https://en.cppreference.com/
- **(new)** *Effective Modern C++* (Scott Meyers) · CppCon talks — https://www.youtube.com/@CppCon **Stage T2 — DSA in C++, sustained (~200+ hrs):** ~4–6 problems/week, spaced, pattern-based, then timed.
- NeetCode 150 — https://neetcode.io/ · LeetCode — https://leetcode.com/ · MIT 6.006 — https://ocw.mit.edu/courses/6-006-introduction-to-algorithms-spring-2020/ · Abdul Bari — https://www.youtube.com/@abdul_bari · CLRS
- **(new)** Grokking the Coding Interview / DesignGurus — https://www.designgurus.io/ · **Striver's A2Z DSA + SDE sheet** (excellent, India-friendly) — https://takeuforward.org/ · Sean Prashad's LeetCode Patterns — https://seanprashad.com/leetcode-patterns/ · *Competitive Programmer's Handbook* (free) — https://cses.fi/book/book.pdf · CSES Problem Set — https://cses.fi/problemset/

Checkpoint (rolling): implement from memory in C++ the core structures + standard patterns (two-pointer, sliding window, binary search, backtracking, DP, graphs); solve a random LeetCode medium in ~25–30 min and a hard in ~40–45.

Thread M — Implement-by-hand ML · ~3 hrs/week *(up from v6's 2)* · highest-ROI thread + concept-cementer

Validated: DeepMind's RE coding round asks you to implement components/losses from a blank file (e.g. "implement Focal Loss from scratch in PyTorch" — your own domain loss). For you it does double duty — interview skill AND cementing concepts (implementing something is the fastest way to truly understand it). ~3 hrs/wk ≈ 150–170 hrs. From a **blank file, no autocomplete, no AI**, reimplement one primitive in ~30–45 min, then revisit older ones spaced; later, add a second weekly "build a small system end-to-end" slot.

Rotating list: backprop through an MLP · scaled dot-product + multi-head attention · a segmentation loss (Dice/Focal/Tversky) · cross-entropy from logits · LayerNorm/BatchNorm · softmax + stable log-softmax · a sampler (top-k/nucleus/temperature) · a DDPM step · logistic/linear regression + gradients · k-means · PCA via SVD · a tiny SGD/Adam loop · positional encodings (sinusoidal/RoPE) · a KV-cache · a message-passing GNN layer · a small VQE circuit · translate primitives to JAX.

- d2l.ai https://d2l.ai/ · labml.ai https://nn.labml.ai/ · Annotated Transformer http://nlp.seas.harvard.edu/annotated-transformer/ · micrograd https://github.com/karpathy/micrograd · nanoGPT https://github.com/karpathy/nanoGPT · target rounds https://igotanoffer.com/en/advice/google-deepmind-research-engineer-interview
- **(new)** eriklindernoren/ML-From-Scratch — https://github.com/eriklindernoren/ML-From-Scratch · Karpathy nn-zero-to-hero repo — https://github.com/karpathy/nn-zero-to-hero · minGPT — https://github.com/karpathy/minGPT · tinygrad (read the source) — https://github.com/tinygrad/tinygrad · Raschka *Build a Large Language Model (from Scratch)* — https://github.com/rasbt/LLMs-from-scratch

Thread R — Research output, networking & applications · ~2 hrs/week *(up from v6's 1.5)*

The Tier-2/PhD differentiator and your pipeline engine. Absorbs old habits #6 (reproduce-a-paper) and #7 (positioning doc). ~2 hrs/wk ≈ 100–110 hrs.

- **Reproduce-a-target-paper (monthly):** publicly reproduce one paper from a target team and write it up. Code: https://paperswithcode.com/
- **(new) Read papers well:** "How to Read a Paper" (Keshav) — https://web.stanford.edu/class/ee384m/Handouts/HowtoReadPaper.pdf · Connected Papers — https://www.connectedpapers.com/ · Papers We Love — https://github.com/papers-we-love/papers-we-love
- **Write:** 1–2 short posts/month; maintain the "why me for digital biology" narrative + two job-talk decks.
- **Publish:** aim the spike capstone at a venue — MICCAI https://www.miccai.org/ · MIDL https://midl.io/ · MLSB https://www.mlsb.io/ · LoG https://logconference.org/ · ISMB/RECOMB https://www.iscb.org/ · arXiv https://arxiv.org/ · bioRxiv https://www.biorxiv.org/ · OpenReview https://openreview.net/
- **Applications:** run the company ladder (below); from Tier 0 this thread carries your live loops.

BLOCK 1 — Foundations, domain & the spike

PHASE 0 — Rapid Brush-Up → Fundamentals Rebuild: Stats, ML & DL

~55 hrs *(v6 45)* · genuine rebuild + terminology glossary + spoken command · do this first · cures the articulation pain point · after this you start applying

Goal: rebuild crisp, *out-loud* command of the fundamentals and vocabulary. Go thoroughly, intuition-first, and **build your living glossary as you go**. Deliverable: fluent spoken articulation + a glossary you trust.

Primary — StatQuest: https://www.youtube.com/@statquest/playlists · https://statquest.org/

- **0.1 Statistics fundamentals (~14 hrs)** — distributions; mean/variance/SD; populations vs samples; CLT; sampling; hypothesis testing & p-values; CIs; R^2; covariance & correlation; MLE; bootstrapping; power. *Verbal:* p-value, CI, CLT, MLE each in two sentences, no notes.
- **0.2 Classic ML (~20 hrs)** — bias–variance; train/val/test & CV; confusion matrix, sensitivity/specificity, precision/recall, ROC & AUC (and when

PR is honest); linear & logistic regression; ridge vs lasso and why L1→sparsity; trees/RF/AdaBoost/GBM/XGBoost; SVMs & the kernel trick; k-means & hierarchical clustering; PCA; naive Bayes; entropy & mutual info. *Verbal, cold:* GBM vs RF; why L1 zeros weights and L2 shrinks (at the gradient level); what AUC measures and when PR-AUC is more honest.

- **0.3 Deep-learning fundamentals (~14 hrs)** — StatQuest NN + 3Blue1Brown NN https://www.youtube.com/playlist?list=PLZHQObOWTQDNU6R1_67000Dx_ZCJB-3pi
- **0.4 Deliverable — articulation cheat-sheet + glossary v1 (~7 hrs).**
- **(new) reference:** StatQuest *The StatQuest Illustrated Guide to Machine Learning* (book) — for the intuition-first picture.

Phase 0 self-test (Feynman gate): explain 15 of the above out loud, plain language, 2–3 sentences each, no notes.

PHASE 1 — Foundations: Mathematics + Neural Networks from First Principles

~210 hrs *(v6 190)* · the bedrock · a genuine build — do NOT rush · intuition-first throughout

Goal: real mathematical maturity — the thing that lets you push a model past its vanilla ceiling and explain *why* it works. **Visual explainer → rigorous lecture → derive by hand**, with spaced re-derivation of anything slippery.

- **1.1 Linear algebra (~72 hrs) — highest-leverage topic.**
  - 3B1B Essence of LA — https://www.youtube.com/playlist?list=PLZHQObOWTQDPD3MizzM2xVFitgF8hE_ab · MIT 18.06 (Strang) — https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/video_galleries/video-lectures/ · MIT 18.065 — https://ocw.mit.edu/courses/18-065-matrix-methods-in-data-analysis-signal-processing-and-machine-learning-spring-2018/ · Matrix Cookbook — https://www.math.uwaterloo.ca/~hwolkowi/matrixcookbook.pdf
  - **(new)** Strang *Introduction to Linear Algebra* (book) · Imperial College "Mathematics for ML: Linear Algebra" (Coursera) — https://www.coursera.org/learn/linear-algebra-machine-learning
  - *Whiteboard (spaced):* derive SVD from eigendecomposition; rank-r approximation geometrically; A^TA is PSD; backprop through a linear layer with matrix calculus.
- **1.2 Calculus + matrix calculus (~26 hrs).**
  - 3B1B Essence of Calculus — https://www.youtube.com/playlist?list=PLZHQObOWTQDMsr9K-rj53DwVRMYO3t5Yr · Matrix Calculus for DL (Parr & Howard) — https://explained.ai/matrix-calculus/
  - **(new)** Imperial "Mathematics for ML: Multivariate Calculus" (Coursera) — https://www.coursera.org/learn/multivariate-calculus-machine-learning
- **1.3 Probability & statistics (~50 hrs).**
  - Harvard Stat 110 (Blitzstein) — https://projects.iq.harvard.edu/stat110/youtube · MIT RES.6-012 (Tsitsiklis) — https://ocw.mit.edu/courses/res-6-012-introduction-to-probability-spring-2018/
  - **(new)** Blitzstein & Hwang *Introduction to Probability* (free) — http://probabilitybook.net/
  - *Whiteboard:* MLE for a Gaussian; conditional expectation as projection; Bayes on a non-trivial example.
- **1.4 Neural networks from first principles (~62 hrs) — the bridge · PROTECT.** All 10 videos with exercises, re-implementing each without looking at his code.
  - Karpathy — Neural Networks: Zero to Hero — https://www.youtube.com/playlist?list=PLAqhIrjkxbuWI23v9cThsA9GvCAUhRvKZ · hub https://karpathy.ai/zero-to-hero.html · repo https://github.com/karpathy/nn-zero-to-hero

Phase 1 self-test (gate → Threads T, M, R start): hand-derive backprop for a 2-layer MLP; implement micrograd-style autodiff from memory; explain why BatchNorm helps and what breaks without it; whiteboard scaled dot-product attention and derive the $\sqrt{d_k}$ scaling.

PHASE 2 — Applied Computer Vision & Medical Image Segmentation

~195 hrs *(v6 185)* · your domain, at full v6 depth + added practice · ends with Capstone 1 → Tier 0/1 · apply in parallel

Goal: turn what you *do* into what you can *explain and improve* — conv math, loss math, "why it works" — plus the public portfolio piece. (All v6 sub-blocks retained at full hours; SEQUENCING OPTION: you may begin applying to Tier-0 India roles during this phase.)

- **2.1 Imaging physics (~30 hrs).** Elster Q&A — https://mriquestions.com/index.html · OHBM — https://www.youtube.com/@ohbmonline/playlists · Callaghan (first 4) — https://www.youtube.com/playlist?list=PLqEMVgX2VgZcG_xR1QRdsuc-X_zEgMx-Z · iBiology fluorescence — https://www.youtube.com/playlist?list=PLF513KEDjY9YBNhwMfX6jMI3PlpdW9TMP
- **2.2 CNNs + convolution math + U-Net family + losses (~50 hrs).** CS231n 2017 — https://www.youtube.com/playlist?list=PL3FW7Lu3i5JvHM8ljYj-zLfQRF3EO8sYv · https://cs231n.github.io/ · Conv arithmetic — https://arxiv.org/abs/1603.07285 · https://github.com/vdumoulin/conv_arithmetic · U-Net https://arxiv.org/abs/1505.04597 · V-Net https://arxiv.org/abs/1606.04797 · Attention U-Net https://arxiv.org/abs/1804.03999 · nnU-Net https://www.nature.com/articles/s41592-020-01008-z · Losses — survey https://arxiv.org/abs/2006.14822 · Generalized Dice https://arxiv.org/abs/1707.03237 · Focal Tversky https://arxiv.org/abs/1810.07842 · Boundary loss https://arxiv.org/abs/1812.07032 (implement each from scratch — Thread M).
- **2.3 nnU-Net + MONAI + microscopy hands-on (~45 hrs).** MONAI — https://github.com/Project-MONAI/tutorials · videos https://www.youtube.com/@projectmonai/videos · nnU-Net v2 — https://github.com/MIC-DKFZ/nnUNet · MONAI integration https://github.com/Project-MONAI/tutorials/blob/main/nnunet/README.md · Kitware https://www.kitware.com/developing-custom-3d-medical-image-segmentation-solutions-using-out-of-the-box-pipelines-in-monai/ · DigitalSreeni — https://www.youtube.com/@DigitalSreeni/playlists · code https://github.com/bnsreenu/python_for_microscopists · AI for Medical Diagnosis — https://www.coursera.org/learn/ai-for-medical-diagnosis
- **2.4 Transformer-based segmentation + foundation models (~20 hrs).** ViT https://arxiv.org/abs/2010.11929 · TransUNet https://arxiv.org/abs/2102.04306 · Swin-UNet https://arxiv.org/abs/2105.05537 · UNETR https://arxiv.org/abs/2103.10504 · Swin UNETR https://arxiv.org/abs/2201.01266 · SAM https://arxiv.org/abs/2304.02643 · MedSAM https://www.nature.com/articles/s41467-024-44824-z · MedNeXt https://arxiv.org/abs/2303.09975
- **2.5 3D, uncertainty, evaluation (~40 hrs).** TorchIO — https://torchio.readthedocs.io/ · Uncertainty (Yarin Gal) — https://www.youtube.com/results?search_query=yarin+gal+uncertainty+deep+learning · Metrics Reloaded — https://www.nature.com/articles/s41592-023-02151-z · Stanford AIMI — https://aimi.stanford.edu/education/educational-resources
  - *3D-conv exercise:* input (D=64,H=128,W=128,C=4), kernel (3,3,3), stride (1,2,2), padding (1,1,1), 32 out-channels — output shape + param count by hand; repeat for transposed + dilated.

PHASE 2 — CAPSTONE 1 (~25 hrs) · public data, COI-safe · Tier-0/1 gate. Notebook + 3-page write-up: preprocessing (physics-justified), architecture ablation (3 architectures), loss ablation, uncertainty + failure-mode analysis. Implement losses + one architecture block from scratch.

- **Datasets:** Medical Decathlon http://medicaldecathlon.com/ · BraTS https://www.synapse.org/brats · TCIA https://www.cancerimagingarchive.net/ · IXI https://brain-development.org/ixi-dataset/ · LIVECell https://github.com/sartorius-research/LIVECell · Sartorius https://www.kaggle.com/competitions/sartorius-cell-instance-segmentation · BBBC https://bbbc.broadinstitute.org/

Phase 2 self-test: whiteboard U-Net forward pass + skips with shapes; why nnU-Net usually beats fancier architectures (and when not); diagnose bias-field artifacts, class imbalance, label noise, domain shift.

PHASE 3 — Deep Learning Theory & Modern Architectures

~165 hrs *(v6 150)* · the modeling-fundamentals core · deepened, intuition-first, triangulated

Goal: build real modeling intuition — optimization dynamics, information theory, modern architectures, generative/diffusion theory. Where "why does my model behave this way, and how do I make it better" is answered.

- **3.1 Optimization for deep learning (~30 hrs).** Ruder — https://www.ruder.io/optimizing-gradient-descent/ · Boyd EE364A — https://www.youtube.com/playlist?list=PL3940DD956CDF0622 · book https://web.stanford.edu/~boyd/cvxbook/
- **3.2 Information theory for ML (~20 hrs).** MacKay ITILA ch 1–6 — https://www.inference.org.uk/itila/book.html
- **3.3 Transformers in depth + modern variants (~44 hrs).** CS25 — https://web.stanford.edu/class/cs25/ · playlist https://www.youtube.com/playlist?list=PLoROMvodv4rNiJRchCzutFw5ItR_Z27CM · Lilian Weng — https://lilianweng.github.io/ · Annotated Transformer — http://nlp.seas.harvard.edu/annotated-transformer/ · RoPE https://arxiv.org/abs/2104.09864 · RMSNorm https://arxiv.org/abs/1910.07467 · FlashAttention https://arxiv.org/abs/2205.14135 · GQA/MQA https://arxiv.org/abs/2305.13245 · SwiGLU https://arxiv.org/abs/2002.05202 · MoE/Switch https://arxiv.org/abs/2101.03961
- **3.4 Vision transformers + modern ConvNets (~27 hrs).** ViT lectures (Lucas Beyer) — https://www.youtube.com/results?search_query=lucas+beyer+vision+transformer · ConvNeXt https://arxiv.org/abs/2201.03545 · DINOv2 https://arxiv.org/abs/2304.07193 · MAE https://arxiv.org/abs/2111.06377 · HF CV course — https://huggingface.co/learn/computer-vision-course
- **3.5 Generative modeling & diffusion (~44 hrs) · PROTECT — feeds the spike.** CS236 — https://www.youtube.com/playlist?list=PLoROMvodv4rPOWA-omMM6STXaWW4FvJT8 · DDPM https://arxiv.org/abs/2006.11239 · Yang Song https://yang-song.net/blog/2021/score/ · Lilian Weng diffusion https://lilianweng.github.io/posts/2021-07-11-diffusion-models/ · HF Diffusion (units 1–2) — https://huggingface.co/learn/diffusion-course

Phase 3 self-test: derive the diffusion loss (variational + score-matching); compare positional encodings; explain SSL in low-data regimes; explain what changes training dynamics with momentum/warmup/schedules. **→ Tier 1 (broad).**

PHASE 5 — Full-Stack / Production AI Systems

~165 hrs *(v6 160)* · production breadth + ML-systems-design (in the RE loop)

(Numbered 5 to preserve v6 alignment; Phase Q sits between 4/6 conceptually — see below.)

- **5.1 LLM internals (~34 hrs).** CS336 — https://stanford-cs336.github.io/spring2024/ · playlist https://www.youtube.com/playlist?list=PLoROMvodv4rOY23Y0BoGoBGgQ1zmU_MT_ · HF NLP — https://huggingface.co/learn/nlp-course
- **5.2 RAG at senior level (~28 hrs).** Hamel Husain — https://www.oreilly.com/radar/what-we-learned-from-a-year-of-building-with-llms-part-i/ · Eugene Yan patterns — https://eugeneyan.com/writing/llm-patterns/ · Anthropic contextual retrieval — https://www.anthropic.com/news/contextual-retrieval · LlamaIndex https://docs.llamaindex.ai/ · RAGAS https://docs.ragas.io/ · DSPy https://dspy.ai/
- **5.3 MLOps + ML systems design (~44 hrs) · KEEP.** Chip Huyen — https://huyenchip.com/ml-interviews-book/contents/8.1.1-ml-systems-design.html · CS329S — https://stanford-cs329s.github.io/ · Made With ML — https://madewithml.com/ · Full Stack DL — https://fullstackdeeplearning.com/course/2022/
- **5.4 Distributed training + inference optimization (~26 hrs).** Raschka — https://magazine.sebastianraschka.com/ · HF multi-GPU — https://huggingface.co/docs/transformers/en/perf_train_gpu_many · vLLM https://arxiv.org/abs/2309.06180 · quantization (GPTQ/AWQ/GGUF/FP8) · ZeRO https://arxiv.org/abs/1910.02054
- **5.5 Eval / observability / safety + ML-systems-design prep (~33 hrs).** Anthropic Constitutional AI — https://www.anthropic.com/news/constitutional-ai · Eugene Yan evals — https://eugeneyan.com/writing/llm-evaluators/ · observability (LangSmith / Phoenix / Helicone); *ML-systems-design* whiteboards (RAG for HCPs; brain-seg pipeline with domain-shift handling; drug-discovery digital-twin retraining + monitoring).

PHASE 6 — Geometric / Molecular ML — THE SPIKE

~185 hrs *(v6 150 — deepened)* · the differentiator for the relocation target + the PhD seed · ends with Capstone 3 (flagship) → Tier 2

Goal: close the geometric-DL / molecular-AI gap that connects all your work — the highest-reward phase for the relocation target. Your growing QML understanding (Phase Q) makes the "quantum-informed" angle uniquely credible, on 100% public benchmarks.

- **6.1 Geometric deep learning + graph ML (~50 hrs).** CS224W — https://web.stanford.edu/class/cs224w/ · 2023 https://snap.stanford.edu/class/cs224w-2023/ · playlist https://www.youtube.com/playlist?list=PLoROMvodv4rPLKxIpqhjhPgdQy7imNkDn · Bronstein GDL — https://geometricdeeplearning.com/lectures/ · proto-book https://arxiv.org/abs/2104.13478 · Veličković https://www.youtube.com/results?search_query=petar+velickovic+gnn+lectures · PyG https://pytorch-geometric.readthedocs.io/ · DGL https://www.dgl.ai/
  - **(new)** PyG "Learn" / colabs — https://pytorch-geometric.readthedocs.io/en/latest/get_started/colabs.html
- **6.2 Molecular ML + equivariant networks (~55 hrs) · the core.** DeepChem — https://deepchem.io/tutorials/ · TDC https://tdcommons.ai/ · SchNet https://arxiv.org/abs/1706.08566 · DimeNet++ https://arxiv.org/abs/2011.14115 · GemNet https://arxiv.org/abs/2106.08903 · MolFormer https://arxiv.org/abs/2106.09553 · Cohen & Welling https://arxiv.org/abs/1602.07576 · EGNN https://arxiv.org/abs/2102.09844 · e3nn https://e3nn.org/ · https://github.com/e3nn/e3nn · tutorial https://blondegeek.github.io/e3nn_tutorial/ · NequIP https://github.com/mir-group/nequip · paper https://www.nature.com/articles/s41467-022-29939-5 · MACE https://github.com/ACEsuit/mace · paper https://arxiv.org/abs/2206.07697 · Allegro https://github.com/mir-group/allegro · Equiformer https://arxiv.org/abs/2206.11990 · curated list https://github.com/Eipgen/Neural-Netw

ork-Models-for-Chemistry

- **6.3 Protein structure + equivariance (~27 hrs).** AlphaFold 2 (AlQuraishi) — https://www.youtube.com/results?search_query=AlphaFold+2+explained+AlQuraishi · paper https://www.nature.com/articles/s41586-021-03819-2 · ESM/ESMFold https://github.com/facebookresearch/esm · ESM Atlas https://esmatlas.com/ · Boltz https://github.com/jwohlwend/boltz · OpenFold https://github.com/aqlaboratory/openfold
- **6.4 JAX + mock interviews + portfolio (~28 hrs).** Recorded, reviewed: 45-min segmentation deep-dive; 30-min behavioral. **Translate a capstone component to JAX** — https://jax.readthedocs.io/ · Equinox https://docs.kidger.site/equinox/ · **(new)** Flax — https://flax.readthedocs.io/ .

PHASE 6 — CAPSTONE 3 (~35 hrs, flagship) · public quantum-chemistry data, COI-safe · unlocks Tier 2 · the Isomorphic-differentiator piece. Train an **E(3)-equivariant GNN** (SchNet → DimeNet++ → EGNN/NequIP-style) to predict molecular properties; ablate equivariance; **fold in the quantum-informed angle** (delta-ML toward CCSD; DFT-vs-quantum data-generation — connects to Phase Q). Implement the message-passing core from scratch; translate a component to JAX.

- **Datasets:** QM9 https://quantum-machine.org/datasets/ · PCQM4Mv2 https://ogb.stanford.edu/kddcup2021/pcqm4m/ · MoleculeNet https://moleculenet.org/ · OGB https://ogb.stanford.edu/ · TDC https://tdcommons.ai/
- **Angle:** portfolio + an MLSB / AI4Science / LoG workshop paper. **The flagship — do it well.**

Phase 6 self-test: message passing vs convolution; what equivariance buys and why it matters for molecules; when a GNN beats a fingerprint baseline; how AlphaFold's structure module uses equivariance; the quantum-chemistry data-generation story end-to-end.

PHASE Q — Quantum Computing & QML (full-depth track)

~305 hrs *(v6 275 + capstone 30)* · a genuine build from near-zero — you want the expertise · intuition-first · + added resources

Your quantum exposure was overseeing a vendor (QpiAI's retrosynthesis work), not hands-on depth, and you want to become good at QML — so this is a complete build at full v6 hours, with new resources added. ***Two sequencing options — you choose:*** **(A) Integrated (v6-style):** run Phase Q inside Block 1 (between Phases 5 and 6, as v6 did), so everything finishes ~March–May 2027 — heavier weekly load, but quantum is "done" on the v6 timeline. **(B) Parallel/after-Tier-2 (default):** run it parallel-lite (a few hrs/week) or as a concentrated block after you reach Tier-2 — you interview sooner and give quantum deep, uninterrupted focus later (better for a slow-intuitive learner). Full depth either way. It seeds a QML career branch (QpiAI) and deepens the molecular spike.

Intuition-first entry (before the rigorous courses): Quantum Country (spaced-repetition primer) — https://quantum.country/qcvc · MIT 8.04 (Adams) — https://ocw.mit.edu/courses/8-04-quantum-physics-i-spring-2013/video_galleries/lecture-videos/

- **Q.1 Group theory & symmetry (~22 hrs) — shared with the spike's equivariance.** Group theory for physicists — https://www.youtube.com/results?search_query=group+theory+for+physicists+lectures · GDL lectures — https://geometricdeeplearning.com/lectures/
- **Q.2 QM math: Hilbert spaces, operators, tensor products (~25 hrs).** MIT 8.04 — https://ocw.mit.edu/courses/8-04-quantum-physics-i-spring-2013/ · Quantum Country — https://quantum.country/qcvc · (optional) Schuller — https://www.youtube.com/playlist?list=PLPH7f_7ZlzxTi6kS4vCmv4ZKm9u8g5yic
- **Q.3 Quantum computing fundamentals — Watrous / IBM track (~80 hrs) · rigorous core.** Basics — https://learning.quantum.ibm.com/course/basics-of-quantum-information · YouTube https://www.youtube.com/playlist?list=PLOFEBzvs-VvrXTMy5Y2IqmSaUjfnhvBHR · Algorithms — https://learning.quantum.ibm.com/course/fundamentals-of-quantum-algorithms · General Formulation — https://learning.quantum.ibm.com/course/general-formulation-of-quantum-information · Error Correction — https://learning.quantum.ibm.com/course/foundations-of-quantum-error-correction · Nielsen & Chuang ch 1–4, 8, 10
  - **(new)** Qiskit textbook / IBM Quantum Learning hub — https://learning.quantum.ibm.com/ · John Preskill's Caltech Ph219 notes — http://theory.caltech.edu/~preskill/ph219/ · Xanadu/PennyLane Codebook — https://pennylane.ai/codebook · IBM Quantum Challenges (hands-on) — https://www.ibm.com/quantum
- **Q.4 Variational quantum algorithms — VQE, ADAPT-VQE, DMET, QAOA (~45 hrs).** PennyLane quantum-chemistry demos — https://pennylane.ai/qml/demos_quantum-chemistry · VQE https://www.nature.com/articles/ncomms5213 · ADAPT-VQE https://www.nature.com/articles/s41467-019-10988-2 · DMET-VQE https://arxiv.org/abs/2108.08987 · QAOA https://arxiv.org/abs/1411.4028 · Hadfield https://arxiv.org/abs/1709.03489 · Pauli grouping https://arxiv.org/abs/1907.13117
- **Q.5 Quantum machine learning theory (~42 hrs) — the honest-take material.** PennyLane QML (Schuld-curated) — https://pennylane.ai/topics/quantum-machine-learning · Schuld "Taking stock of QML" — https://www.youtube.com/results?search_query=maria+schuld+taking+stock+quantum+machine+learning · Schuld & Petruccione ch 4–7 · QML=kernels https://arxiv.org/abs/2101.11020 · Barren plateaus https://www.nature.com/articles/s41467-018-07090-4 · Expressibility https://arxiv.org/abs/1905.10876 · Advantage-goal https://arxiv.org/abs/2203.01340 · PQC encodings https://arxiv.org/abs/2008.08605 · Péré https://github.com/Christophe-pere/QML-Course · Hjorth-Jensen https://github.com/CompPhysics/QuantumComputingMachineLearning
  - **(new)** TensorFlow Quantum — https://www.tensorflow.org/quantum · PennyLane demos gallery — https://pennylane.ai/qml/demonstrations/
- **Q.6 Quantum chemistry on quantum computers + retrosynthesis context (~60 hrs).** Ties to what you oversaw at QpiAI. Qiskit Global Summer School — https://www.youtube.com/@qiskit/playlists · IBM Quantum Learning + Sample-based/Krylov Quantum Diagonalization — https://learning.quantum.ibm.com/ · https://www.ibm.com/quantum/blog/iql-migration

PHASE Q — CAPSTONE Q (~30 hrs) · public molecules, COI-safe. Take a **public** molecule (H_2/LiH/H_2O or a QM9 subset — **not** Lilly molecules, and clear of the QpiAI engagement specifics). Re-implement three ways — vanilla VQE + UCCSD, ADAPT-VQE, DMET-VQE — benchmark against classical CCSD; write-up defending every choice (ansatz, optimizer, measurement grouping, noise mitigation, basis set) + a barren-plateau analysis. Optionally add a classical-ML surrogate (delta-ML toward CCSD) — the "quantum × ML" story connecting to Capstone 3.

Phase Q self-test: derive VQE from the variational principle; parameter-shift rule and why it works; the barren-plateau argument; VQE vs QAOA — when each and when *no* quantum algorithm; an honest take on QML hype vs promise (cite Schuld); DMET fragmentation and the classical-quantum interface. *(Feynman gate.)*

BLOCK 2 — Breadth, CV-Expert & Research/Lab-Interview Readiness

~560 hrs + interviewing · Threads T + M + R continue underneath

PHASE 7 — Generative AI & Agentic AI (domain-focused)

~155 hrs *(v6 150)* · competent, domain-focused · ends with Capstone 4

- **7.1 Advanced LLMs — adaptation & inference (~42 hrs).** LoRA https://arxiv.org/abs/2106.09685 · QLoRA https://arxiv.org/abs/2305.14314 · Raschka LoRA https://magazine.sebastianraschka.com/ · InstructGPT/RLHF https://arxiv.org/abs/2203.02155 · DPO https://arxiv.org/abs/2305.18290 · HF TRL · HF LLM course https://huggingface.co/learn/llm-course · Chip Huyen *AI Engineering* (2025)
- **7.2 Multimodal & vision-language models (~26 hrs).** CLIP https://arxiv.org/abs/2103.00020 · LLaVA https://arxiv.org/abs/2304.08485 · Flamingo https://arxiv.org/abs/2204.14198
- **7.3 Agents & agentic systems (~52 hrs).** HF Agents Course https://huggingface.co/learn/agents-course · HF MCP Course https://huggingface.co/learn/mcp-course · Anthropic Building Effective Agents https://www.anthropic.com/research/building-effective-agents · Andrew Ng patterns https://www.deeplearning.ai/short-courses/ · LangGraph https://langchain-ai.github.io/langgraph/ · patterns: ReAct, tool/function calling, reflection, planning, memory, multi-agent, agent eval.

PHASE 7 — CAPSTONE 4 (~35 hrs) · public literature, COI-safe. A multi-step research/analysis agent (RAG + tool use + memory + eval) over **open** scientific literature.

- **Corpora/tools:** PMC-OA https://www.ncbi.nlm.nih.gov/pmc/tools/openftlist/ · arXiv https://arxiv.org/ · PubMed E-utilities https://www.ncbi.nlm.nih.gov/books/NBK25501/ · Semantic Scholar API https://www.semanticscholar.org/product/api. **Never ingest Lilly internal docs/data.**

Phase 7 self-test: explain LoRA and why it's parameter-efficient; RLHF vs DPO; when an agent beats a single well-prompted call; walk the ReAct loop; how you'd evaluate agent reliability. **→ Tier 2 broadened.**

PHASE 8 — Computer Vision Expert (+ Video + C++/edge) — *restored to full v6 + new resources*

~245 hrs *(v6 230)* · full CV-expert breadth so no vision question catches you flat · ends with Capstone 2 (video) + Capstone 5 (deploy)

Goal: round out into a broad CV expert — classical & geometric vision, 3D, video/temporal, detection/tracking — plus the deployment/optimization skills that put CV models onto real hardware. **All v6 sub-blocks retained at full hours, with new resources/practice added.** *SEQUENCING OPTION: pull 8.3–8.4 (video/temporal + life-sciences video) forward to right after Phase 2 to unlock Tier-1 imaging roles (Recursion) earlier — hours stay the same, only the order shifts.*

- **8.1 Classical & geometric CV (~45 hrs).** First Principles of CV (Nayar) — https://fpcv.cs.columbia.edu/ · Coursera https://www.coursera.org/specializations/firstprinciplesofcomputervision · Cyrill Stachniss — https://www.youtube.com/@CyrillStachniss · Hartley & Zisserman *Multiple View Geometry*
  - **(new)** Szeliski *Computer Vision: Algorithms and Applications* (free) — https://szeliski.org/Book/ · COLMAP (SfM) — https://colmap.github.io/ · OpenMVG — https://github.com/openMVG/openMVG · ORB-SLAM3 — https://github.com/UZ-SLAMLab/ORB_SLAM3 · Open3D — http://www.open3d.org/
- **8.2 Modern DL for vision — breadth (~35 hrs).** Michigan EECS 498 (Justin Johnson) — https://web.eecs.umich.edu/~justincj/teaching/eecs498/ · playlist https://www.youtube.com/playlist?list=PL5-TkQAfAZFbzxjBHtzdVCWE0Zbhomg7r
  - **(new)** detection/segmentation frameworks for practice — Detectron2 https://github.com/facebookresearch/detectron2 · MMDetection https://github.com/open-mmlab/mmdetection · Ultralytics YOLO https://docs.ultralytics.com/
- **8.3 Video & temporal analytics (~40 hrs).** VideoMAE https://arxiv.org/abs/2203.12602 · TimeSformer https://arxiv.org/abs/2102.05095 · ViViT https://arxiv.org/abs/2103.15691 · SlowFast https://arxiv.org/abs/1812.03982 · SAM 2 https://arxiv.org/abs/2408.00714 · https://github.com/facebookresearch/sam2 · RAFT https://arxiv.org/abs/2003.12039 · https://github.com/princeton-vl/RAFT · ByteTrack https://arxiv.org/abs/2110.06864 · https://github.com/ifzhang/ByteTrack · OC-SORT https://arxiv.org/abs/2203.14360 · https://github.com/noahcao/OC_SORT · BoT-SORT https://arxiv.org/abs/2206.14651 · https://github.com/NirAharon/BoT-SORT · HOTA https://arxiv.org/abs/2009.07736 · MOTChallenge https://motchallenge.net/ · DAVIS https://davischallenge.org/ · YouTube-VOS https://youtube-vos.org/
- **8.4 Life-sciences-native video (~30 hrs) — the differentiator.** DeepLabCut https://www.deeplabcut.org/ · https://github.com/DeepLabCut/DeepLabCut · Nature Neuroscience https://www.nature.com/articles/s41593-018-0209-y · SLEAP https://sleap.ai/ · https://github.com/talmolab/sleap · Nature Methods https://www.nature.com/articles/s41592-022-01426-1 · Cellpose https://www.cellpose.org/ · https://github.com/MouseLand/cellpose · TrackMate https://imagej.net/plugins/trackmate/ · Ultrack https://github.com/royerlab/ultrack · Cell Tracking Challenge http://celltrackingchallenge.net/
- **8.5 Deployment & optimization for hardware/edge (~50 hrs).** OpenCV in C++ — https://docs.opencv.org/ · applied recipes https://pyimagesearch.com/ · ONNX https://onnx.ai/ · TensorRT https://developer.nvidia.com/tensorrt · CUDA basics (NVIDIA intro)
  - **(new)** *Programming Massively Parallel Processors* (Kirk & Hwu) · NVIDIA DLI CUDA course — https://www.nvidia.com/en-us/training/ · Triton Inference Server — https://github.com/triton-inference-server/server · NVIDIA DeepStream — https://developer.nvidia.com/deepstream-sdk · CUDA C++ Programming Guide — https://docs.nvidia.com/cuda/cuda-c-programming-guide/
- **8.6 3D vision & neural rendering (~30 hrs).** NeRF https://arxiv.org/abs/2003.08934 · 3D Gaussian Splatting https://arxiv.org/abs/2308.04079 · depth estimation, point-cloud processing
  - **(new)** nerfstudio — https://docs.nerf.studio/ · Open3D point-cloud tutorials — http://www.open3d.org/docs/release/tutorial/geometry/pointcloud.html

PHASE 8 — CAPSTONE 2 (video, ~included) · public data, COI-safe · Tier 1. DeepLabCut/SLEAP pose → temporal model → behavior classification; **or** Cellpose seg → TrackMate/Ultrack tracking + lineage; SAM 2 for mask propagation.

- **Datasets:** CalMS21 https://data.caltech.edu/records/s0vdx-0k302 · MABe https://www.aicrowd.com/challenges/multi-agent-behavior-challenge-2022 · Cell Tracking Challenge http://celltrackingchallenge.net/

PHASE 8 — CAPSTONE 5 (deployment, ~included) · public data, COI-safe · restored. A geometric-CV or 3D task deployed with a C++/optimized inference component: camera calibration + SfM on public sets, feature-based visual odometry, **or** take C1/C2's model and get it running optimized (TensorRT/ONNX, quantized) behind a C++ path.

- **Datasets:** KITTI https://www.cvlibs.net/datasets/kitti/ · TUM RGB-D https://cvg.cit.tum.de/data/datasets/rgbd-dataset · ETH3D https://www.eth3 d.net/

Phase 8 self-test: derive the pinhole camera model + epipolar constraint; SIFT and why scale/rotation invariance matters; two-stage vs one-stage detectors; SAM 2's memory mechanism; tracking-by-detection vs joint detection-and-tracking; what TensorRT does; how you'd port a PyTorch CV model to a real-time C++ pipeline.

PHASE 9 — Research + Big-Tech Interview Crescendo

~160 hrs *(v6 150)* · peak the interview skills right as you interview · recalibrated to the RE loop · unlocks Tier 3

RE loop = **2 medium-hard coding rounds + ML fundamentals + ML system design + behavioral**, plus a research discussion for research-adjacent roles. This is where the robust year-long threads convert into performance.

- **9.1 DSA — RE weight (~55 hrs) — crescendo of Thread T.** Timed NeetCode 150 → Blind 75 → company-tagged — https://neetcode.io/ · https://leetcode.com/ ; patterns under pressure — DesignGurus https://www.designgurus.io/ · Striver SDE sheet https://takeuforward.org/ . Out loud, blank editor, ~35 min, **code must run** (CoderPad-style); ~4–6/day this phase.
- **9.2 Implement-by-hand ML (~35 hrs) — crescendo of Thread M · highest value.** Timed, unaided, from a blank file: attention, a loss (Dice/Focal), a sampler, a diffusion step, a training loop, backprop, a message-passing layer. Plus an ML math/theory sweep (gradient computation, KL properties, sampling, convergence intuition, the "L1 sparsity via gradients" mechanism — asked verbatim).
- **9.3 System design (~35 hrs) — distinct from Phase 5.** *ML* system design first (training/serving/data/eval/monitoring, domain shift); ByteByteGo — https://bytebytego.com/ · https://www.youtube.com/@ByteByteGo · Gaurav Sen https://www.youtube.com/@gkcs · DDIA (Kleppmann) + MIT 6.824 https://pdos.csail.mit.edu/6.824/ for the loops that include generic system design. Whiteboard 8–10 prompts, recorded, self-critiqued.
- **9.4 Research job-talk + behavioral (~35 hrs).** A 20-min talk + Q&A on each of your two strongest capstones (segmentation/video + the molecular spike). ~6–8 STAR stories per org's signals; low-level/OO design (parking lot, elevator, rate limiter) for loops that include it; live peer mocks — Pramp / interviewing.io. **Research loops ban AI tools — rehearse fully unaided.**

Phase 9 self-test: a random LeetCode hard in ~45 min with clean, running code; implement attention + a loss + a sampler + a GNN layer from a blank file, unaided, timed; a full system-design whiteboard in 45 min; a crisp 20-min job-talk; four STAR stories from memory. **→ Tier 3.**

(Active interviewing: keep all three threads warm with ~5 hrs/week maintenance; buffer for on-sites, take-homes, rest.)

Application overlay & company target ladder

*Your applied medical-imaging skill makes you eligible at Tier 0 **now** — apply after ~1–2 months of fundamentals brush-up and climb as each capstone lands. Match the tier; early loops pay you in feedback. "Relocate" flags roles outside India.*

Tier	Unlocked by	~When	Apply to
Tier 0	Applied skill + Capstone 1	~Nov–Dec 2026 (apply from month ~2)	India medical-imaging AI + accessible; stretch Google-India
Tier 1	Capstone 1 + Capstone 2 (imaging + video)	~Jan–Feb 2027	broader medical/biomedical imaging + video; stronger Google-India case
Tier 2	Capstone 3 (molecular GNN, flagship)	~Mar–Apr 2027	AI drug-discovery labs; **Isomorphic / DeepMind-science RE realistic**
Tier 3	Interview crescendo (P9)	~Jul–Aug 2027	full top-lab loops
QML branch	Capstone Q (Phase Q)	when Phase Q completes	quantum-for-science roles (below)

Tier 0 — India-domestic + accessible: Qure.ai https://qure.ai/ · SigTuple (microscopy/cell — closest to your fluorescence work) https://sigtuple.com/ · NVIDIA healthcare/MONAI + CUDA/TensorRT https://www.nvidia.com/en-us/about-nvidia/careers/ · GE HealthCare (Bengaluru R&D) https://www.gehealthcare.com/ · Microsoft Research India https://www.microsoft.com/en-us/research/lab/microsoft-research-india/ · Google Research India https://research.google/locations/india/

Tier 1 — broaden: Aidoc, Rad AI, Annalise.ai, PathAI, Paige.AI, Owkin (Paris/NYC) https://www.owkin.com/

Tier 2 — AI drug discovery + research labs (relocate): Recursion (*phenomics = deep learning on cell images — a direct match*) https://www.recursion.com/careers · Insitro https://insitro.com/ · Genesis Therapeutics https://www.genesistherapeutics.ai/ · Iambic https://www.iambic.ai/ · Chai Discovery https://www.chaidiscovery.com/ · EvolutionaryScale https://www.evolutionaryscale.ai/ · Profluent https://www.profluent.bio/ · InstaDeep https://www.instadeep.com/ · Valence Labs https://www.valencelabs.com/ · **Isomorphic Labs** https://www.isomorphiclabs.com/careers · **Google DeepMind** https://deepmind.google/about/careers/

- *Target **Research Engineer / ML Research Engineer** postings* (accept MSc + ~2 yrs ML experience — no PhD). Save Research Scientist for after a PhD. · *Isomorphic has a Lilly partnership — not a blocker, apply with normal discretion.*

QML career branch — quantum + AI for science (leverages Phase Q + your QpiAI history): QpiAI (Bengaluru; QpiAI-Pharma for drug discovery; quantum-chemistry / quantum-ML / AI-native roles) https://www.qpiai.tech/careers · IBM Quantum https://www.ibm.com/quantum · quantum-for-chemistry / materials teams broadly.

Tier 3 — top research loops: Isomorphic, Google DeepMind, Google Research (global + India), Genentech gRED / Prescient Design, Microsoft Research Health Futures, Meta FAIR.

Conflict-of-interest guardrails

Not legal advice — verify specifics with the right people at Lilly.

1. **Public data only.** Public benchmarks (Medical Decathlon/BraTS, QM9, PCQM4Mv2, CalMS21, PMC-OA, KITTI, etc.). Never Lilly proprietary data, targets, molecules, assays, or internal docs. In Capstone Q/Capstone 3 use textbook/public molecules (H_2/LiH/H_2O/QM9), **not** Lilly molecules (avoid the keto-enol / HATU work), and **clear of anything specific to the QpiAI retrosynthesis engagement** you oversaw.
2. **Generic, published methods** in any portfolio artifact — no Lilly (or vendor) trade secrets.
3. **Personal time and resources** — your own hardware/accounts; not Lilly compute, licenses, or work hours. *(Work on Lilly hardware/networks must use Lilly's JFrog Artifactory for packages per enterprise policy; these capstones are explicitly personal-hardware/personal-account.)*
4. **Clear before you publish** — Lilly's standard external-publication / IP-review process; when unsure, ask first.
5. **Check your agreement** — employment / invention-assignment / moonlighting terms.
6. **Positioning stays generic** — transferable skills, never Lilly (or vendor) specifics.

Cross-cutting habits (every week, no exceptions)

De-duplicated: old habits #3, #6, #7 now live inside Threads M and R.

1. **Paper-of-the-week** — one paper, deeply read, derivations redone on paper.
2. **Whiteboard Friday** — one concept whiteboarded without notes; photograph it → interview-prep deck.
3. **Sunday writeup** — a 1-page note teaching the week's deepest concept to a hypothetical junior.
4. **War chest + living glossary** — one doc for derivations, paper one-liners, gotchas, anti-patterns, and every confusing term in your own words (Confusion Buffer #2).

*(Re-implementation Saturday → **Thread M**. Reproduce-a-target-paper + positioning doc → **Thread R**.)*

PhD-later decision guide

Whether. For the long-term **Senior-RS** aspiration, a PhD is effectively required at the target labs (Isomorphic RS asks for PhD + publications; RE accepts MSc + ~2 yrs). So: RS title → PhD; senior/staff RE → PhD optional. *Your Phase-Q QML depth + the molecular spike also open a **quantum-ML / computational-chemistry PhD**, a distinct well-funded niche.*

When. *After* landing a strong RE role — job-first expands options.

Where (and how job-first improves each):

- **India — part-time / external (moonlight-compatible).** IISc's **External Registration Programme** and the IITs' external/part-time PhD tracks (≈2 yrs' experience; IISc needs ~1 yr on-campus + a company co-advisor; part-time ~6 yrs). — https://www.iisc.ac.in/admissions/external-registration-programme-ph-d/
- **EU — salaried / industrial (a career move).** NL/Germany/Sweden/Switzerland/Austria PhDs are salaried employee roles (~€3–6k/mo, ~4 yrs); MSCA Industrial Doctoral Networks embed the PhD across a company + university. The strongest prestige-and-funding leap. — https://euraxess.ec.europa.eu/
- **USA — mostly full-time funded; part-time rare.** More realistic as a research-embedded path, or skip it via the RE→staff ladder.

Set-up now (zero marginal cost): the spike capstone → a non-archival workshop paper + a bioRxiv/arXiv preprint; 1–2 advisor relationships; Capstone Q gives a second (QML) research identity if you want the quantum-PhD route.

Decision criteria (priority order): advisor fit → topic continuity → funding model (salaried/industrial > self-funded) → part-time feasibility → location/visa → prestige (last).

Publication & portfolio strategy

Realistic output for a working professional over ~18–24 months: 1–2 non-archival workshop papers + 1 preprint + clean open-source repos.

- **Job lever.** Reproduce-a-paper + a workshop paper = paper-discussion prep, referral hook, proof of level. MLSB prioritizes **open-sourced code** for spotlights — the code *is* the credential.
- **PhD-optionality lever.** First-author workshop papers + preprints + an advisor relationship lift you from "self-funded tier-2/3 PhD" to "funded EU/industrial or research-embedded."
- **Venues & timing.** Imaging/video → MIDL / MICCAI workshop (spring/summer) https://midl.io/ · https://www.miccai.org/ . Molecular → MLSB (NeurIPS, ~Sep/Oct submission, Dec) https://www.mlsb.io/ · LoG https://logconference.org/ · AI4Science / Simbiochem (NeurIPS). QML → a quantum-ML workshop or arXiv. Target **MIDL-2027 / MLSB-2027 + a rolling bioRxiv/arXiv preprint**.
- **Visibility:** open-source every capstone with a reproducible README; a merged PR into an ecosystem repo (**e3nn / MACE / PyG / MONAI / PennyLane**) beats another blog post.

Risk register

Risk	Likelihood	Impact	Leading indicator	Mitigation
Confusion compounding (shaky foundations carried forward) — *your key risk*	High	High	Self-tests failed/skipped; a term keeps reappearing unexplained	The **Confusion Buffer** — understanding-gated checkpoints, spaced re-derivation, consolidation weeks, multi-teacher triangulation. Don't advance until the Feynman gate passes
Overload / burnout (v9 is	Hig	Hi	Skipped rest/consolidation	Steady sustainable weekly load; threads + consolidation weeks are the shock

~2,390 hrs + robust threads)	h	gh	weeks; capstone quality dropping; dread	absorbers; protect one rest day; the timeline is pre-approved to extend rather than compress
Breadth-without-depth (esp. with full CV-expert + full QML)	Med	High	Can *use* but can't derive/teach a topic	Self-tests are the gate; Thread M + Whiteboard Friday enforce depth; the spike stays the priority for the relocation target
QML pulling focus off the job path	Med	Med	Doing Phase Q before Tier-2 without a QML-role target	Default to option B (parallel/after-Tier-2); pull earlier only for a QpiAI-type target
Over-claiming CV seniority	Med	High	Positioning says "senior CV engineer"; freeze on classical-vision/SLAM	You now *have* the depth (full Phase 8) — but still position as "medical-imaging + geometric/molecular ML" primary, CV breadth as strong supporting
Goal drift / re-bloating	Med	Med	Adding "just one more" course	Every addition serves imaging-video **or** molecular-geometric **or** the RE loop **or** (deliberately) QML/CV-breadth — else cut
Relocation-vs-domestic split	Med	Med	Effort split 50/50, mediocre at both	Two legs share an umbrella; apply Tier-0 India *now* while building the Tier-2 spike
Premature / ill-fitting PhD	Low–Med	High	Applying to a self-funded tier-2/3 PhD before landing the job	PhD is post-job; job-first expands options

Resource index (quick lookup) — *all v6/v7/v8 URLs preserved; (new) = added in v9*

Brush-up (P0) — StatQuest https://www.youtube.com/@statquest/playlists · https://statquest.org/ · 3B1B NN https://www.youtube.com/playlist?list=PLZHQObOWTQDNU6R1_67000Dx_ZCJB-3pi

Foundations math (P1) — 3B1B LA https://www.youtube.com/playlist?list=PLZHQObOWTQDPD3MizzM2xVFitgF8hE_ab · 3B1B Calc https://www.youtube.com/playlist?list=PLZHQObOWTQDMsr9K-rj53DwVRMYO3t5Yr · MIT 18.06 https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/ · MIT 18.065 https://ocw.mit.edu/courses/18-065-matrix-methods-in-data-analysis-signal-processing-and-machine-learning-spring-2018/ · Stat 110 https://projects.iq.harvard.edu/stat110/youtube · MIT RES.6-012 https://ocw.mit.edu/courses/res-6-012-introduction-to-probability-spring-2018/ · Matrix Cookbook https://www.math.uwaterloo.ca/~hwolkowi/matrixcookbook.pdf · Matrix Calculus https://explained.ai/matrix-calculus/ · Karpathy Z2H https://www.youtube.com/playlist?list=PLAqhIrjkxbuWI23v9cThsA9GvCAUhRvKZ · hub https://karpathy.ai/zero-to-hero.html · **(new)** Imperial Math-for-ML LA https://www.coursera.org/learn/linear-algebra-machine-learning · Multivariate Calculus https://www.coursera.org/learn/multivariate-calculus-machine-learning · Blitzstein & Hwang Probability http://probabilitybook.net/ · Z2H repo https://github.com/karpathy/nn-zero-to-hero

Later-threaded math — Boyd https://www.youtube.com/playlist?list=PL3940DD956CDF0622 · https://web.stanford.edu/~boyd/cvxbook/ · Ruder https://www.ruder.io/optimizing-gradient-descent/ · MacKay https://www.inference.org.uk/itila/book.html · Group theory https://www.youtube.com/results?search_query=group+theory+for+physicists+lectures · GDL lectures https://geometricdeeplearning.com/lectures/

Medical imaging (P2) — MRI Q&A https://mriquestions.com/index.html · OHBM https://www.youtube.com/@ohbmonline/playlists · Callaghan https://www.youtube.com/playlist?list=PLqEMVgX2VgZcG_xR1QRdsuc-X_zEgMx-Z · iBiology https://www.youtube.com/playlist?list=PLF513KEDjY9YBNhwMfX6jMl3PlpdW9TMP · AI for Medical Diagnosis https://www.coursera.org/learn/ai-for-medical-diagnosis · DigitalSreeni https://www.youtube.com/@DigitalSreeni/playlists · code https://github.com/bnsreenu/python_for_microscopists · Conv arithmetic https://arxiv.org/abs/1603.07285 · https://github.com/vdumoulin/conv_arithmetic · MONAI https://github.com/Project-MONAI/tutorials · https://www.youtube.com/@projectmonai/videos · nnU-Net https://github.com/MIC-DKFZ/nnUNet · TorchIO https://torchio.readthedocs.io/ · Stanford AIMI https://aimi.stanford.edu/education/educational-resources · CS231n https://www.youtube.com/playlist?list=PL3FW7Lu3i5JvHM8ljYj-zLfQRF3EO8sYv · https://cs231n.github.io/ · Metrics Reloaded https://www.nature.com/articles/s41592-023-02151-z · U-Net https://arxiv.org/abs/1505.04597 · V-Net https://arxiv.org/abs/1606.04797 · Attention U-Net https://arxiv.org/abs/1804.03999 · nnU-Net https://www.nature.com/articles/s41592-020-01008-z · Loss survey https://arxiv.org/abs/2006.14822 · Generalized Dice https://arxiv.org/abs/1707.03237 · Focal Tversky https://arxiv.org/abs/1810.07842 · Boundary loss https://arxiv.org/abs/1812.07032 · ViT https://arxiv.org/abs/2010.11929 · TransUNet https://arxiv.org/abs/2102.04306 · Swin-UNet https://arxiv.org/abs/2105.05537 · UNETR https://arxiv.org/abs/2103.10504 · Swin UNETR https://arxiv.org/abs/2201.01266 · SAM https://arxiv.org/abs/2304.02643 · MedSAM https://www.nature.com/articles/s41467-024-44824-z · MedNeXt https://arxiv.org/abs/2303.09975 · **Datasets:** Medical Decathlon http://medicaldecathlon.com/ · BraTS https://www.synapse.org/brats · TCIA https://www.cancerimagingarchive.net/ · IXI https://brain-development.org/ixi-dataset/ · LIVECell https://github.com/sartorius-research/LIVECell · Sartorius https://www.kaggle.com/competitions/sartorius-cell-instance-segmentation · BBBC https://bbbc.broadinstitute.org/

Deep learning core (P3) — CS25 https://web.stanford.edu/class/cs25/ · https://www.youtube.com/playlist?list=PLoROMvodv4rNiJRchCzutFw5ItR_Z27CM · CS336 https://stanford-cs336.github.io/spring2024/ · https://www.youtube.com/playlist?list=PLoROMvodv4rOY23Y0BoGoBGgQ1zmU_MT_ · CS236 https://www.youtube.com/playlist?list=PLoROMvodv4rPOWA-omMM6STXaWW4FvJT8 · Lilian Weng https://lilianweng.github.io/ · Annotated Transformer http://nlp.seas.harvard.edu/annotated-transformer/ · HF NLP https://huggingface.co/learn/nlp-course · HF CV https://huggingface.co/learn/computer-vision-course · HF Diffusion https://huggingface.co/learn/diffusion-course · RoPE https://arxiv.org/abs/2104.09864 · RMSNorm https://arxiv.org/abs/1910.07467 · FlashAttention https://arxiv.org/abs/2205.14135 · GQA https://arxiv.org/abs/2305.13245 · SwiGLU https://arxiv.org/abs/2002.05202 · Switch/MoE https://arxiv.org/abs/2101.03961 · ConvNeXt https://arxiv.org/abs/2201.03545 · DINOv2 https://arxiv.org/abs/2304.07193 · MAE https://arxiv.org/abs/2111.06377 · DDPM https://arxiv.org/abs/2006.11239 · Score-based https://yang-song.net/blog/2021/score/ · Diffusion (blog) https://lilianweng.github.io/posts/2021-07-11-diffusion-models/

Quantum & QML (Phase Q) — IBM Quantum Learning https://learning.quantum.ibm.com/ · Watrous https://www.youtube.com/playlist?list=PLOFEBzvs-VvrXTMy5Y2IqmSaUjfnhvBHR · Basics https://learning.quantum.ibm.com/course/basics-of-quantum-information · Algorithms https://learning.quantum.ibm.com/course/fundamentals-of-quantum-algorithms · General Formulation https://learning.quantum.ibm.com/course/general-formulation-of-quantum-information · Error Correction https://learning.quantum.ibm.com/course/foundations-of-quantum-error-correction · Qiskit https://www.youtube.com/@qiskit/playlists · QDA blog https://www.ibm.com/quantum/blog/iql-migration · PennyLane codebook https://pennylane.ai/codebook · PennyLane QML https://pennylane.ai/topics/quantum-machine-learning · PennyLane chemistry https://pennylane.ai/qml/demos_quantum-chemi stry · MIT 8.04 https://ocw.mit.edu/courses/8-04-quantum-physics-i-spring-2013/ · Quantum Country https://quantum.country/qcvc · Schuller https://www.youtube.com/playlist?list=PLPH7f_7ZlzxTi6kS4vCmv4ZKm9u8g5yic · Péré https://github.com/Christophe-pere/QML-Course · Hjorth-Jensen https://github.com/CompPhysics/QuantumComputingMachineLearning · VQE https://www.nature.com/articles/ncomms5213 · ADAPT-VQE https://www.nature.com/articles/s41467-019-10988-2 · DMET-VQE https://arxiv.org/abs/2108.08987 · QAOA https://arxiv.org/abs/1411.4028 · Hadfield https://arxiv.org/abs/1709.03489 · QML=kernels https://arxiv.org/abs/2101.11020 · Barren plateaus https://www.nature.com/articles/s41467-018-07090-4 · Expressibility https://arxiv.org/abs/1905.10876 · Advantage-goal https://arxiv.org/abs/2203.01340 · PQC encodings https://arxiv.org/abs/2008.08605 · Pauli grouping https://arxiv.org/abs/1907.13117 · **(new)** Preskill Ph219 http://theory.caltech.edu/~preskill/ph219/ · TensorFlow Quantum https://www.tensorflow.org/quantum · PennyLane demos gallery https://pennylane.ai/qml/demonstrations/ · IBM Quantum https://www.ibm.com/quantum

Production / MLOps (P5) — CS329S https://stanford-cs329s.github.io/ · Made With ML https://madewithml.com/ · Full Stack DL https://fullstackdeeplearning.com/course/2022/ · Chip Huyen https://huyenchip.com/ml-interviews-book/ · Hamel Husain https://www.oreilly.com/radar/what-we-learned-from-a-year-of-building-with-llms-part-i/ · Eugene Yan patterns https://eugeneyan.com/writing/llm-patterns/ · Eugene Yan evals https://eugeneyan.com/writing/llm-evaluators/ · Anthropic contextual retrieval https://www.anthropic.com/news/contextual-retrieval · Anthropic Constitutional AI https://www.anthropic.com/news/constitutional-ai · LlamaIndex https://docs.llamaindex.ai/ · RAGAS https://docs.ragas.io/ · DSPy https://dspy.ai/ · Raschka https://magazine.sebastianraschka.com/ · HF multi-GPU https://huggingface.co/docs/transformers/en/perf_train_gpu_many · vLLM https://arxiv.org/abs/2309.06180 · ZeRO https://arxiv.org/abs/1910.02054

DSA & C++ (Thread T) — learncpp https://www.learncpp.com/ · The Cherno https://www.youtube.com/@TheCherno · cppreference https://en.cppreference.com/ · NeetCode https://neetcode.io/ · LeetCode https://leetcode.com/ · MIT 6.006 https://ocw.mit.edu/courses/6-006-introduction-to-algorithms-spring-2020/ · Abdul Bari https://www.youtube.com/@abdul_bari · **(new)** DesignGurus https://www.designgurus.io/ · Striver takeuforward https://takeuforward.org/ · Sean Prashad patterns https://seanprashad.com/leetcode-patterns/ · CP Handbook https://cses.fi/book/book.pdf · CSES https://cses.fi/problemset/ · CppCon https://www.youtube.com/@CppCon

Implement-by-hand ML (Thread M) — d2l.ai https://d2l.ai/ · labml.ai https://nn.labml.ai/ · Annotated Transformer http://nlp.seas.harvard.edu/annotated-transformer/ · micrograd https://github.com/karpathy/micrograd · nanoGPT https://github.com/karpathy/nanoGPT · DeepMind RE round guide https://igotanoffer.com/en/advice/google-deepmind-research-engineer-interview · JAX https://jax.readthedocs.io/ · Equinox https://docs.kidger.site/equinox/ · Anki https://apps.ankiweb.net/ · **(new)** ML-From-Scratch https://github.com/eriklindernoren/ML-From-Scratch · nn-zero-to-hero https://github.com/karpathy/nn-zero-to-hero · minGPT https://github.com/karpathy/minGPT · tinygrad https://github.com/tinygrad/tinygrad · LLMs-from-scratch (Raschka) https://github.com/rasbt/LLMs-from-scratch · Flax https://flax.readthedocs.io/

Research output (Thread R) — Papers with Code https://paperswithcode.com/ · MICCAI https://www.miccai.org/ · MIDL https://midl.io/ · MLSB https://www.mlsb.io/ · LoG https://logconference.org/ · ISCB/ISMB https://www.iscb.org/ · arXiv https://arxiv.org/ · bioRxiv https://www.biorxiv.org/ · OpenReview https://openreview.net/ · **(new)** How to Read a Paper (Keshav) https://web.stanford.edu/class/ee384m/Handouts/HowtoReadPaper.pdf · Connected Papers https://www.connectedpapers.com/ · Papers We Love https://github.com/papers-we-love/papers-we-love

GenAI & Agents (P7) — HF LLM course https://huggingface.co/learn/llm-course · HF Agents https://huggingface.co/learn/agents-course · HF MCP https://huggingface.co/learn/mcp-course · Anthropic Building Effective Agents https://www.anthropic.com/research/building-effective-agents · DeepLearning.AI short courses https://www.deeplearning.ai/short-courses/ · LangGraph https://langchain-ai.github.io/langgraph/ · LoRA https://arxiv.org/abs/2106.09685 · QLoRA https://arxiv.org/abs/2305.14314 · InstructGPT/RLHF https://arxiv.org/abs/2203.02155 · DPO https://arxiv.org/abs/2305.18290 · CLIP https://arxiv.org/abs/2103.00020 · LLaVA https://arxiv.org/abs/2304.08485 · Flamingo https://arxiv.org/abs/2204.14198 · **Corpora:** PMC-OA https://www.ncbi.nlm.nih.gov/pmc/tools/openftlist/ · PubMed E-utilities https://www.ncbi.nlm.nih.gov/books/NBK25501/ · Semantic Scholar API https://www.semanticscholar.org/product/api

Computer Vision Expert + Video (P8) — First Principles of CV https://fpcv.cs.columbia.edu/ · FPCV Coursera https://www.coursera.org/specializations/firstprinciplesofcomputervision · Cyrill Stachniss https://www.youtube.com/@CyrillStachniss · Michigan EECS 498 https://web.eecs.umich.edu/~justincj/teaching/eecs498/ · playlist https://www.youtube.com/playlist?list=PL5-TkQAfAZFbzxjBHtzdVCWE0Zbhomg7r · NeRF https://arxiv.org/abs/2003.08934 · 3D Gaussian Splatting https://arxiv.org/abs/2308.04079 · OpenCV https://docs.opencv.org/ · PyImageSearch https://pyimagesearch.com/ · ONNX https://onnx.ai/ · TensorRT https://developer.nvidia.com/tensorrt · **Video:** VideoMAE https://arxiv.org/abs/2203.12602 · TimeSformer https://arxiv.org/abs/2102.05095 · ViViT https://arxiv.org/abs/2103.15691 · SlowFast https://arxiv.org/abs/1812.03982 · SAM 2 https://arxiv.org/abs/2408.00714 · https://github.com/facebookresearch/sam2 · RAFT https://arxiv.org/abs/2003.12039 · https://github.com/princeton-vl/RAFT · ByteTrack https://arxiv.org/abs/2110.06864 · https://github.com/ifzhang/ByteTrack · OC-SORT https://arxiv.org/abs/2203.14360 · https://github.com/noahcao/OC_SORT · BoT-SORT https://arxiv.org/abs/2206.14651 · https://github.com/NirAharon/BoT-SORT · HOTA https://arxiv.org/abs/2009.07736 · MOTChallenge https://motchallenge.net/ · DAVIS https://davischallenge.org/ · YouTube-VOS https://youtube-vos.org/ · **Life-sciences video:** DeepLabCut https://www.deeplabcut.org/ · SLEAP https://sleap.ai/ · Cellpose https://www.cellpose.org/ · TrackMate https://imagej.net/plugins/trackmate/ · Ultrack https://github.com/royerlab/ultrack · Cell Tracking Challenge http://celltrackingchallenge.net/ · CalMS21 https://data.caltech.edu/records/s0vdx-0k302 · **Deploy datasets:** KITTI https://www.cvlibs.net/datasets/kitti/ · TUM RGB-D https://cvg.cit.tum.de/data/datasets/rgbd-dataset · ETH3D https://www.eth3d.net/ · **(new)** Szeliski book https://szeliski.org/Book/ · COLMAP https://colmap.github.io/ · OpenMVG https://github.com/openMVG/openMVG · ORB-SLAM3 https://github.com/UZ-SLAMLab/ORB_SLAM3 · Open3D http://www.open3d.org/ · Detectron2 https://github.com/facebookresearch/detectron2 · MMDetection https://github.com/open-mmlab/mmdetection · Ultralytics YOLO https://docs.ultralytics.com/ · nerfstudio https://docs.nerf.studio/ · Triton https://github.com/triton-inference-server/server · DeepStream https://developer.nvidia.com/deepstream-sdk · CUDA Guide https://docs.nvidia.com/cuda/cuda-c-programming-guide/ · NVIDIA DLI https://www.nvidia.com/en-us/training/

Geometric / Molecular ML (P6 — the spike) — CS224W https://web.stanford.edu/class/cs224w/ · https://snap.stanford.edu/class/cs224w-2023/ · playlist https://www.youtube.com/playlist?list=PLoROMvodv4rPLKxIpqhjhPgdQy7imNkDn · Bronstein GDL https://geometricdeeplearning.com/lectures/ · proto-book https://arxiv.org/abs/2104.13478 · PyG https://pytorch-geometric.readthedocs.io/ · colabs https://pytorch-geometric.readthedocs.io/en/latest/get_started/colabs.html · DGL https://www.dgl.ai/ · DeepChem https://deepchem.io/tutorials/ · TDC https://tdcommons.ai/ · SchNet https://arxiv.org/abs/1706.08566 · DimeNet++ https://arxiv.org/abs/2011.14115 · GemNet https://arxiv.org/abs/2106.08903 · MolFormer https://arxiv.org/abs/2106.09553 · Equivariant (Cohen&Welling) https://arxiv.org/abs/1602.07576 · EGNN https://arxiv.org/abs/2102.09844 · e3nn https://e3nn.org/ · https://github.com/e3nn/e3nn · e3nn tutorial https://blondegeek.github.io/e3nn_tutorial/ · NequIP https://github.com/mir-group/nequip · https://www.nature.com/articles/s41467-022-29939-5 · MACE https://github.com/ACEsuit/mace · https://arxiv.org/abs/2206.07697 · Allegro https://github.com/mir-group/allegro · Equiformer https://arxiv.org/abs/2206.11990 · curated list https://github.com/Eipgen/Neural-Network-Models-for-Chemistry · AlphaFold 2 https://www.nature.com/articles/s41586-021-03819-2 · ESM https://github.com/facebookresearch/esm · ESM Atlas https://esmatlas.com/ · Boltz https://github.com/jwohlwend/boltz · OpenFold https://github.com/aqlaboratory/openfold · **Datasets:** QM9 https://quantum-machine.org/datasets/ · PCQM4Mv2 https://ogb.stanford.edu/kddcup2021/pcqm4m/ · MoleculeNet https://moleculenet.org/ · OGB https://ogb.stanford.edu/

Interview (P9) — NeetCode https://neetcode.io/ · LeetCode https://leetcode.com/ · DesignGurus https://www.designgurus.io/ · Striver https://takeuforward.org/ · ByteByteGo https://bytebytego.com/ · https://www.youtube.com/@ByteByteGo · Gaurav Sen https://www.youtube.com/@gkcs · MIT 6.824 https://pdos.csail.mit.edu/6.824/

Company ladder — Qure.ai https://qure.ai/ · SigTuple https://sigtuple.com/ · NVIDIA https://www.nvidia.com/en-us/about-nvidia/careers/ · GE HealthCare https://www.gehealthcare.com/ · MSR India https://www.microsoft.com/en-us/research/lab/microsoft-research-india/ · Google Research India https://research.google/locations/india/ · Owkin https://www.owkin.com/ · Recursion https://www.recursion.com/careers · Insitro https://insitro.com/ · Genesis https://www.genesistherapeutics.ai/ · Iambic https://www.iambic.ai/ · Chai https://www.chaidiscovery.com/ · EvolutionaryScale https://www.evolutionaryscale.ai/ · Profluent https://www.profluent.bio/ · InstaDeep https://www.instadeep.com/ · Valence Labs https://www.valencelabs.com/ · Isomorphic Labs https://www.isomorphiclabs.com/careers · Google DeepMind https://deepmind.google/about/careers/ · **QML branch:** QpiAI https://www.qpiai.tech/careers · IBM Quantum https://www.ibm.com/quantum · **PhD routes:** IISc ERP https://www.iisc.ac.in/admissions/external-registration-programme-ph-d/ · EURAXESS https://euraxess.ec.europa.eu/

A note on what you're taking on

This is a **~2,390-hour** program on top of a full-time role — deliberately robust, built so you never walk into an interview under-confident in *any* area: full math and fundamentals depth (intuition-first, generously timed), the complete CV-expert breadth, a genuine QML build, a protected geometric/molecular spike, and threads sized for real fluency in coding, from-scratch implementation, and research discussion. Anchors: the self-tests and Feynman gates are the true gates, not the calendar — **advance on understanding, not dates**; take consolidation weeks liberally (they're the shock absorber that keeps confusion from compounding); the threads flex down on heavy weeks. **Apply from ~month 2**, on your existing applied imaging skill, and let the incremental ladder carry you from your first India loop upward as each capstone lands. Give **Phase Q** the real hours it deserves *because you want that expertise* — run it parallel/after-Tier-2 by default so it never delays your ladder, or fold it into Block 1 if you'd rather have it done by March. The timeline mirrors v6 (Block 1 core ~March, Block 2 ~July, then expertise grinding), and it's fine for each block to run ~a month longer for the depth — you've chosen mastery over speed, and that's the right call for how you learn. Spend your best focus on the **foundations and the spike**; let breadth and QML compound around them; and remember that the slowness of truly *building* the fundamentals is the point — it's what turns "I use this" into "I understand this well enough to push a model past its ceiling and teach it to a panel."